

APPENDIX C

S-RECORD OUTPUT FORMAT

The S-record format for output modules is for encoding programs or data files in a printable format for transportation between computer systems. The transportation process can be visually monitored, and the S-records can be easily edited.

C.1 S-RECORD CONTENT

Visually, S-records are essentially character strings made of several fields that identify the record type, record length, memory address, code/data, and checksum. Each byte of binary data encodes as a two- character hexadecimal number: the first character represents the high- order four bits, and the second character represents the low-order four bits of the byte. Figure C-1 illustrates the five fields that comprise an S-record. Table C-1 lists the composition of each S- record field.

TYPE	RECORD LENGTH	ADDRESS	CODE/DATA	CHECKSUM
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Figure C-1. Five Fields of an S-Record

Table C-1. Field Composition of an S-Record

Field	Printable Characters	Contents
Type	2	S-record type—S0, S1, etc.
Record Length	2	The count of the character pairs in the record, excluding the type and record length.
Address	4, 6, or 8	The 2-, 3-, or 4-byte address at which the data field is to be loaded into memory.
Code/Data	0–2n	From 0 to n bytes of executable code, memory loadable data, or descriptive information. For compatibility with teletypewriters, some programs may limit the number of bytes to as few as 28 (56 printable characters in the S-record).
Checksum	2	The least significant byte of the one's complement of the sum of the values represented by the pairs of characters making up the record length, address, and the code/data fields.

When downloading S-records, each must be terminated with a CR. Additionally, an S-record may have an initial field that fits other data such as line numbers generated by some time-sharing systems. The record length (byte count) and checksum fields ensure transmission accuracy.

C.2 S-RECORD TYPES

There are eight types of S-records to accommodate the encoding, transportation, and decoding functions. The various Motorola record transportation control programs (e.g. upload, download, etc.), cross assemblers, linkers, and other file creating or debugging programs, only utilize S-records serving the programOs purpose. For more information on support of specific S-records, refer to the userOs manual for that program.

An S-record format module may contain S-records of the following types:

- S0 — The header record for each block of S-records. The code/data field may contain any descriptive information identifying the following block of S-records. Under VERSAdos, the resident linkerOs IDENT command can be used to designate module name, version number, revision number, and description information that will make up the header record. The address field is normally zeros.
- S1 — A record containing code/data and the 2-byte address at which the code/data is to reside.
- S2 — A record containing code/data and the 3-byte address at which the code/data is to reside.
- S3 — A record containing code/data and the 4-byte address at which the code/data is to reside.
- S5 — A record containing the number of S1, S2, and S3 records transmitted in a particular block. This count appears in the address field. There is no code/data field.
- S7 — A termination record for a block of S3 records. The address field may optionally contain the 4-byte address of the instruction to which control is to be passed. There is no code/data field.
- S8 — A termination record for a block of S2 records. The address field may optionally contain the 3-byte address of the instruction to which control is to be passed. There is no code/data field.
- S9 — A termination record for a block of S1 records. The address field may optionally contain the 2-byte address of the instruction to which control is to be passed. Under VERSAdos, the resident linkerOs ENTRY command can be used to specify this address. If this address is not specified, the first entry point specification encountered in the object module input will be used. There is no code/data field.

Each block of S-records uses only one termination record. S7 and S8 records are only active when control is to be passed to a 3- or 4- byte address; otherwise, an S9 is used for termination. Normally, there is only one header record, although it is possible for multiple header records to occur.

C.3 S-RECORD CREATION

Dump utilities, debuggers, a VERSAdos resident linkage editor, or cross assemblers and linkers produce S-record format programs. On VERSAdos systems, the build load module (MBLM) utility allows an executable load module to be built from S-records. It has a counterpart utility in BUILDS that allows an S-record file to be created from a load module.

Programs are available for downloading or uploading a file in S- record format from a host system to an 8- or 16-bit microprocessor- based system. A typical S-record-format module is printed or displayed as follows:

```
S00600004844521B
S1130000285F245F2212226A000424290008237C2A
S11300100002000800082629001853812341001813
S113002041E900084E42234300182342000824A952
S107003000144ED492
S9030000FC
```

The module has an S0 record, four S1 records, and an S9 record. The following character pairs comprise the S-record-format module.

S0 Record:

- S0 — S-record type S0, indicating that it is a header record.
- 06 — Hexadecimal 06 (decimal 6), indicating that six character pairs (or ASCII bytes) follow.
- 0000—A 4-character, 2-byte address field; zeros in this example.
- 48 — ASCII H
- 44 — ASCII D
- 52 — ASCII R
- 1B — The checksum.

First S1 Record:

- S1 — S-record type S1, indicating that it is a code/data record to be loaded/verified at a 2-byte address.
- 13 — Hexadecimal 13 (decimal 19), indicating that 19 character pairs, representing 19 bytes of binary data, follow.
- 0000—A 4-character, 2-byte address field (hexadecimal address 0000) indicating where the data that follows is to be loaded.

The next 16 character pairs of the first S1 record are the ASCII bytes of the actual program code/data. In this assembly language example, the programOs hexadecimal opcodes are sequentially written in the code/data fields of the S1 records.

Opcode	Instruction	
285F	MOVE.L	(A7) +, A4
245F	MOVE.L	(A7) +, A2
2212	MOVE.L	(A2), D1
226A0004	MOVE.L	4(A2), A1
24290008	MOVE.L	FUNCTION(A1), D2
237C	MOVE.L	#FORCEFUNC, FUNCTION(A1)

The rest of this code continues in the remaining S1 recordOs code/data fields and stores in memory location 0010, etc.

2A — The checksum of the first S1 record.

The second and third S1 records also contain hexadecimal 13 (decimal 19) character pairs and end with checksums 13 and 52, respectively. The fourth S1 record contains 07 character pairs and has a checksum of 92.

S9 Record:

- S9 — S-record type S9, indicating that it is a termination record.
- 03 — Hexadecimal 03, indicating that three character pairs (3 bytes) follow.
- 0000—The address field, zeros.
- FC — The checksum of the S9 record.

Each printable character in an S-record encodes in hexadecimal (ASCII in this example) representation of the binary bits that transmit. Figure C-2 illustrates the sending of the first S1 record. Table C-2 lists the ASCII code for S-records.

TYPE				RECORD LENGTH				ADDRESS				CODE/DATA				CHECKSUM			
S	1			1	3			0	0	0	0	2	8	5	F	****	2	A	
5	3	3	1	3	1	3	3	3	0	3	0	3	2	3	8	3	5	4	6
0101	0011	0011	0001	0011	0001	0011	0011	0011	0000	0011	0000	0011	0000	0011	0000	0011	0010	0011	1000
0011	0101	0100	0110	0000	0000	0000	0000	0000	0000	0000	0000	0000	0000	0000	0000	0000	0000	0000	0000

Figure C-2. Transmission of an S1 Record

Table C-2. ASCII Code

Least Significant Digit	Most Significant Digit							
	0	1	2	3	4	5	6	7
0	NUL	DLE	SP	0	@	P	'	p
1	SOH	DC1	!	1	A	Q	a	q
2	STX	DC2	"	2	B	R	b	r
3	ETX	DC3	#	3	C	S	c	s
4	EOT	DC4	\$	4	D	T	d	t
5	ENQ	NAK	%	5	E	U	e	u
6	ACK	SYN	&	6	F	V	f	v
7	BEL	ETB	'	7	G	W	g	w
8	BS	CAN	(	8	H	X	h	x
9	HT	EM	)	9	I	Y	i	y
A	LF	SUB	*	:	J	Z	j	z
B	VT	ESC	+	;	K	[	k	{
C	FF	FS	,	<	L	\	l	
D	CR	GS	-	=	M	]	m	}
E	SO	RS	.	>	N	^	n	~
F	SI	US	/	?	O	_	o	DEL



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